

**4 (short)** The equivalent circuit for a 50 Hz power transformer is shown in Fig. 4.

Under open-circuit conditions,  $\tilde{V}_{out}$  is measured to be 66 kV  $\angle 0$  with  $\tilde{V}_{in} = 11$  kV  $\angle 0$ . The corresponding input current is measured to be  $\tilde{I}_{in} = 0.1$  A at a power factor of 0.8 lagging.

Under short-circuit conditions,  $\tilde{V}_{in}$  is measured to be 0.5 kV  $\angle 0$  with an input current of 3A and an input power of 270 W.

- (a) What is the turns ratio of the transformer? [2]
- (b) Determine the hysteresis and eddy current losses (i.e., those in  $R_0$ ) in the transformer when it is connected to an open circuit. [3]
- (c) Determine the values of the leakage reactance and magnetising reactance of the transformer. In an ideal transformer, what would these values be? [5]

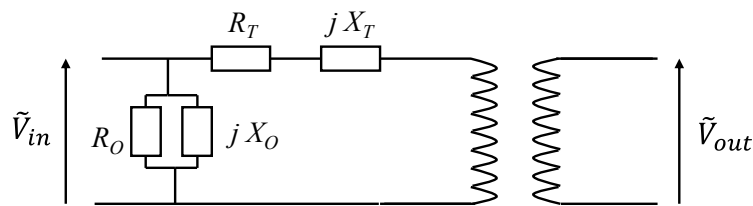


Fig. 4